

# Mohamed Taha Abo Heiba

Cairo, Egypt | +20 114 971 1922 | [mohamed-aboheiba@outlook.com](mailto:mohamed-aboheiba@outlook.com) | Portfolio: [motahaaboheiba.github.io](https://motahaaboheiba.github.io)

LinkedIn: [linkedin.com/in/mohamed-taha-abo-heiba](https://linkedin.com/in/mohamed-taha-abo-heiba) | GitHub: [github.com/MoTahaAboHeiba](https://github.com/MoTahaAboHeiba)

## PROFESSIONAL SUMMARY

**Data Engineer who designs for the failure case, not just the happy path:** schema contracts at ingestion, validated history through SCD2, dimensional models (star, galaxy) built to answer questions no one has asked yet. Shipped across **Databricks, DBT, Airflow, SQL Server, and AWS.** lakehouses, warehouses, and a production RAG. **AWS Certified Cloud Practitioner and Huawei HCIA-Big Data certified, awarded best-performing trainee.**

## EXPERIENCE & TRAINING

- **AWS Cloud Engineer Trainee | National Telecommunication Institute (NTI)** *Oct 2025 – Dec 2025*
  - Architected multi-tier **AWS** app (EC2/RDS/VPC/S3) using **Well-Architected Framework** for fault-tolerant, HA design
  - Ensured high availability via **RDS Multi-AZ** configuration; reduced cold storage costs by implementing S3 lifecycle policies to Glacier
- **Big Data Engineer Trainee | National Telecommunication Institute (NTI)** *Jul 2025 – Sep 2025*
  - Awarded **best-performing trainee**; earned exam voucher and passed Huawei HCIA-Big Data certification on first attempt
  - Built Spark batch pipelines on Hadoop and Kafka, spark streaming across 15 labs, enabling scalable batch and real-time processing
- **Data Engineer Trainee | Digital Egypt Pioneers Initiative (DEPI)** *Apr 2024 – Oct 2024*
  - Automated ETL with 4 SSIS packages that loaded 7 CSVs into SQL Server via 17 procedures, **cutting manual effort**
  - Delivered 70K-record dataset powering churn model at **97.4% accuracy**, enabling targeted at-risk customer retention

## PROJECTS

- **Retail Lakehouse Pipeline [GitHub](#)** | PostgreSQL, Databricks, DBT, Airflow, S3, Docker
  - Built end-to-end **lakehouse**: **Postgres** to **Databricks** Bronze incremental ingestion via **Lakeflow Connect**, **AWS S3** as an external location for secondary ingestion, **dbt**-modeled Silver/Gold layers, **Airflow** (on **Docker**) orchestrating the full chain
  - Modeled Gold as a **Galaxy Schema** with point-in-time **SCD2** resolution across 17 dbt models and 72 passing tests
- **E-Commerce Lakehouse Databricks [GitHub](#)** | Databricks, PySpark, Unity Catalog, Jobs
  - Built an end-to-end **cloud data lakehouse pipeline** processing 200K+ records from 6 data sources through a 3-task Databricks Jobs DAG (**Bronze** ingestion → **Silver** transformation → **Gold** star schema)
  - Delivered a governed, analytics-ready Delta Lake platform using **Medallion Architecture** and **Unity Catalog**, enabling downstream **OLAP consumption** via dimensional **star schema** modeling
- **SQL Data Warehouse Project [GitHub](#)** | SQL Server, T-SQL, Star Schema, Medallion Architecture
  - Architected a fully automated **warehouse pipeline** ingesting 60K+ records **across 6 sources via 2 master stored procedures**, **applying OLTP→OLAP** separation, **star schema** design, and idempotent ETL execution
  - Delivered **star schema** as SQL views with **per-step execution-time logging** for direct bottleneck identification
- **EduMate RAG [Live](#) | [GitHub](#)** | FastAPI, LangChain, Qdrant Cloud, ChromaDB, Huggingface Space
  - Independently built EduMate-RAG, a decoupled microservice (**platform-integrated + standalone modes**) with **hybrid search** over Qdrant Cloud in production and used ChromaDB for dev, indexing 24K+ document chunks via automated PDF pipeline; deployed on Hugging Face Spaces at ~240ms avg latency

## CERTIFICATIONS

**Huawei HCIA Big Data Associate** | Huawei Technologies Co., Ltd

*Issued: Mar 2026 | Valid: Mar 2029*

**AWS Certified Cloud Practitioner (CLF-C02)** | Amazon Web Services

*Issued: May 2026 | Valid: May 2029*

## TECHNICAL SKILLS

- **Data Engineering & ETL Pipelines:** Medallion Architecture, Data Ingestion, (Batch, Stream Processing), Apache Airflow, ETL/ELT
- **Big Data & Distributed Computing:** Spark, Databricks, Lakeflow Connect, Hadoop (HDFS, YARN, Hive), Kafka
- **Data Warehousing:** Dimensional Modeling (Star / Snowflake / Galaxy Schema), T-SQL, Unity Catalog
- **Databases:** SQL Server, PostgreSQL, MySQL, HBase
- **Programming & DevOps:** Python, LangChain, FastAPI, Docker, Git / GitHub
- **Cloud & AWS Data Services:** AWS (S3, EC2, RDS, VPC, IAM, Lambda, Glue, Redshift, ELB, Auto Scaling)
- **Analytics & Visualization:** Power BI, DAX, Power Query

## EDUCATION

**B.Sc. Computer Science — Faculty of Computers & Artificial Intelligence, Fayoum University**

*Oct 2022 – Jun 2026*

GPA: 3.42 / 4.0 | **Core Coursework:** Big Data Engineering, Cloud Computing, Database Management Systems, Advanced SQL, OOP, Data Structures & Algorithms, Machine Learning, Operating Systems (Linux, Docker), Python Programming, Data Visualization (Power BI)